Interpretable Regression Baselines and Diagnostics

SARCOS robot arm inverse dynamics | DS-A3

Research question: how accurately do linear baselines predict first-joint torque from 21 kinematic features, and how does error vary across observed states?

The 44,484 × 28 MATLAB matrix was downloaded from the public Kaixhin/SARCOS GitHub repository. The GPML dataset page documents the data and credits Sethu Vijayakumar. This project does not infer a data license from the repository software license, and the files do not verify physical units.

Target: column 21 (zero-based); predictors: columns 0–20. The original test matrix overlaps the source matrix and was excluded from fitting, selection, and evaluation. Dataset SHA256: b8a249733253ba6097372fedee7696833fcf30de42037d5b4a7227f21a6d1d97

Method: training-mean Dummy, standardized OLS, and standardized Ridge. Scaling is fit within each training fold. Ridge alpha is selected by validation MAE from the preregistered grid; final models are refit on Train + Validation and evaluated once on Test.

Blocked uses contiguous source-row regions with the frozen 25-row boundary purge. Random uses frozen random partitions. Neither establishes episode-independent generalization.

Reproduction: activate nussem1, run the notebook, then run scripts/validate_project.py and python -m unittest discover -s tests -v. Environment: Python: 3.10.18 | NumPy: 2.2.6 | pandas: 2.3.1 | SciPy: 1.15.2 | scikit-learn: 1.7.1

Predictive performance

MAE is the primary metric; RMSE emphasizes larger residuals and R^2 summarizes variance explained. H1 is supported on the tested blocked evaluation: OLS materially improves on the training-mean baseline.
Ridge provides no meaningful improvement in this tested setting. This conclusion is limited to these models, alpha grid, dataset, and evaluation protocols.

Protocol	Partition	Model	MAE	RMSE	R^2
Blocked	Validation	Dummy mean	14.1541	18.9393	-0.0142
Blocked	Validation	OLS	3.6591	5.1754	0.9243
Blocked	Validation	Ridge	3.6591	5.1754	0.9243
Blocked	Test	Dummy mean	11.6312	15.0777	-0.0076
Blocked	Test	OLS	3.8650	5.3396	0.8736
Blocked	Test	Ridge	3.8650	5.3396	0.8736
Random	Validation	Dummy mean	15.3280	20.3378	-0.0004
Random	Validation	OLS	3.9461	5.5460	0.9256
Random	Validation	Ridge	3.9461	5.5460	0.9256
Random	Test	Dummy mean	15.3064	20.2932	-0.0000
Random	Test	OLS	3.9471	5.5700	0.9247
Random	Test	Ridge	3.9471	5.5700	0.9247

Evaluation design and Ridge selection

The blocked protocol holds out contiguous row regions, with the manifest-defined purge gaps; it is sensitive to source ordering and cannot verify episode independence. The random protocol samples held-out rows from the mixed source distribution and may leave nearby observations across partitions.

Blocked validation has markedly lower velocity and acceleration norms than its training and test regions, indicating distribution shift. Random partitions have more similar summaries. The protocol comparison is descriptive and does not by itself establish leakage.

Alpha was selected using Validation MAE only. Test labels did not enter preprocessing, fitting, hyperparameter selection, slice thresholds, or model decisions.

Protocol	Partition	n	Target mean	Median velocity norm	Median acceleration norm
Blocked	Train	26,665	13.3676	2.3018	21.9614
Blocked	Val	8,847	15.6105	0.3636	5.4212
Blocked	Test	8,872	12.6192	2.2242	24.6873
Random	Train	26,690	13.5860	2.0344	19.6477
Random	Val	8,897	13.9716	2.0266	19.4187
Random	Test	8,897	13.5394	2.0118	19.4866

Alpha	Blocked validation MAE	Random validation MAE
0.0010	3.6591	3.9461
0.0100	3.6591	3.9461
0.1000	3.6593	3.9462
1.0000	3.6611	3.9462
10.0000	3.6793	3.9472
100.0000	3.8384	3.9619
1000.0000	4.6976	4.2184

Protocol	Selected alpha
Blocked	0.0010
Random	0.0010

Descriptive error slices

Slices use Blocked Train quartile cutoffs, applied unchanged to Blocked Test. Open-ended end labels preserve the actual digitize assignments; the outside-range count indicates test observations beyond the training min/max.

Higher acceleration and higher absolute torque slices have higher absolute errors. Velocity-slice error is non-monotonic: the low-velocity group has higher MAE than the highest-velocity group. These are descriptive associations. Higher absolute error at large torque does not establish higher relative error or a physical causal mechanism.

Slice variable	Slice interval (train-derived cutoffs)	n	Outside train range count	MAE	RMSE
Velocity norm	(-inf, 1.44069]	2,552	0	4.6970	6.6465
Velocity norm	(1.44069, 2.30176]	2,110	0	3.5896	4.9702
Velocity norm	(2.30176, 2.99481]	2,366	0	3.4456	4.5172
Velocity norm	(2.99481, +inf)	1,844	0	3.5666	4.6465
Acceleration norm	(-inf, 13.8308]	1,922	0	3.1327	4.0621
Acceleration norm	(13.8308, 21.9614]	1,891	0	3.5183	4.6057
Acceleration norm	(21.9614, 31.1808]	2,624	0	4.0954	5.4923
Acceleration norm	(31.1808, +inf)	2,435	0	4.4638	6.4713
Absolute torque	(-inf, 7.11475]	2,210	0	3.2920	4.2341
Absolute torque	(7.11475, 15.3596]	2,541	0	3.1628	4.2628
Absolute torque	(15.3596, 28.7719]	2,779	0	3.8462	5.1743
Absolute torque	(28.7719, +inf)	1,342	0	6.1770	8.3137

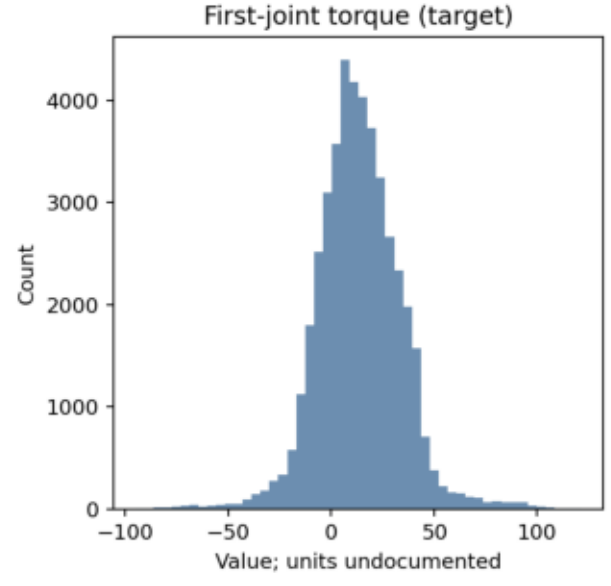
Coefficient summary

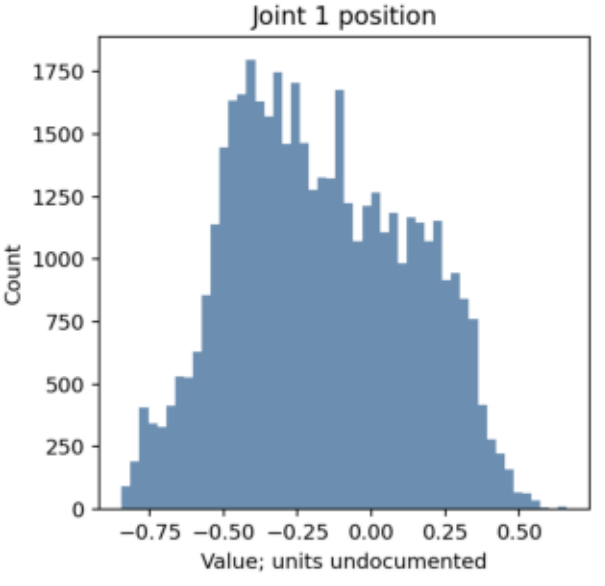
The displayed standardized coefficients are descriptive. Correlated kinematic features limit coefficient-level interpretation; they should not be read as separately identified physical parameters.

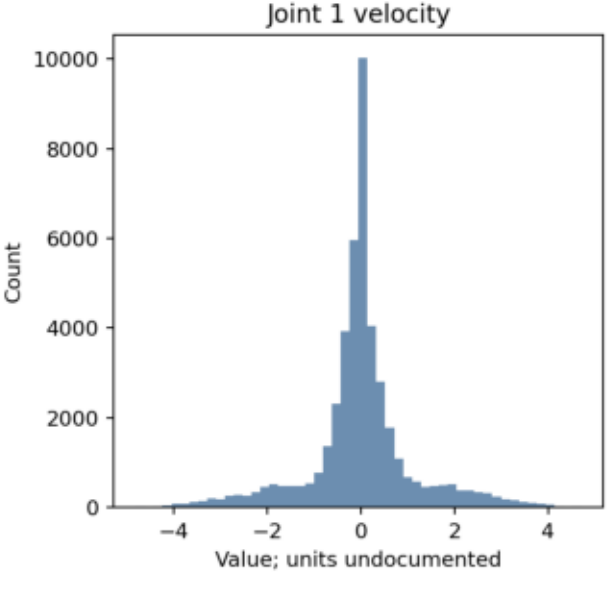
Feature	OLS coef.	Ridge coef.
joint_1_acceleration	26.1780	26.1780
joint_4_acceleration	9.7529	9.7529
joint_1_position	8.7555	8.7555
joint_4_position	-2.7748	-2.7748
joint_5_position	2.0401	2.0401
joint_6_position	1.9672	1.9672
joint_3_acceleration	-1.9135	-1.9135
joint_2_acceleration	-1.7466	-1.7466
joint_5_acceleration	-1.3694	-1.3694
joint_3_position	-1.1819	-1.1819
joint_2_position	0.9444	0.9444
joint_7_position	-0.7501	-0.7501

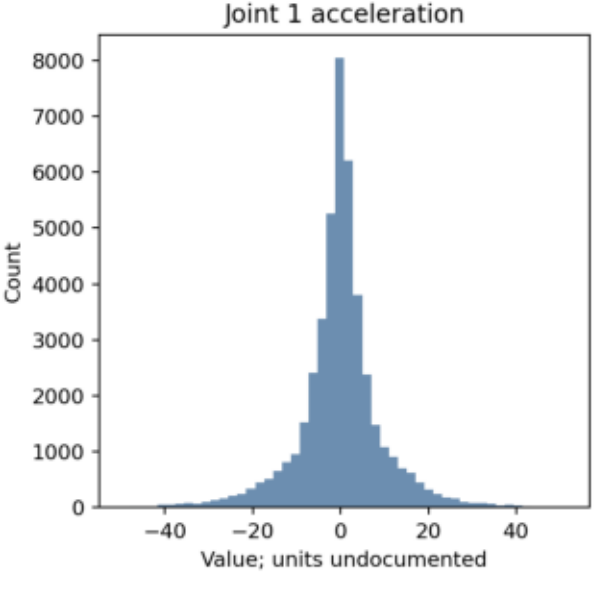
Data exploration

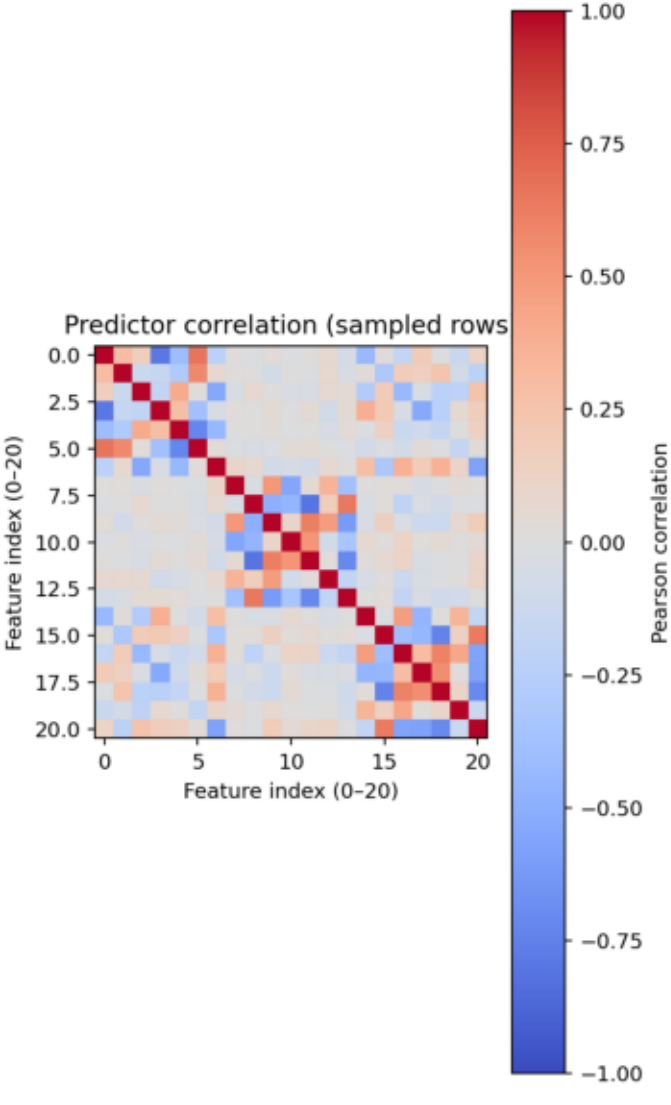
Training data summaries show broad operating-state variation. Marginal distributions do not establish temporal independence or identify physical units.





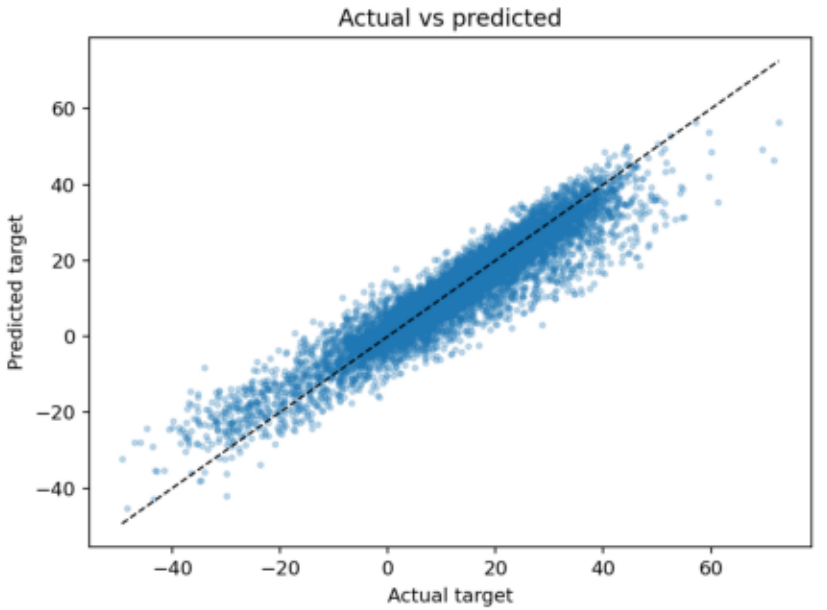


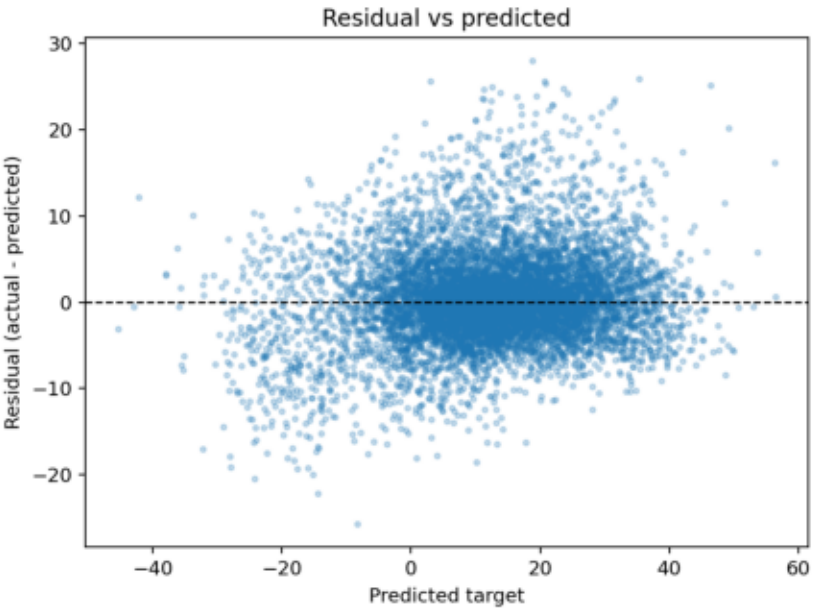


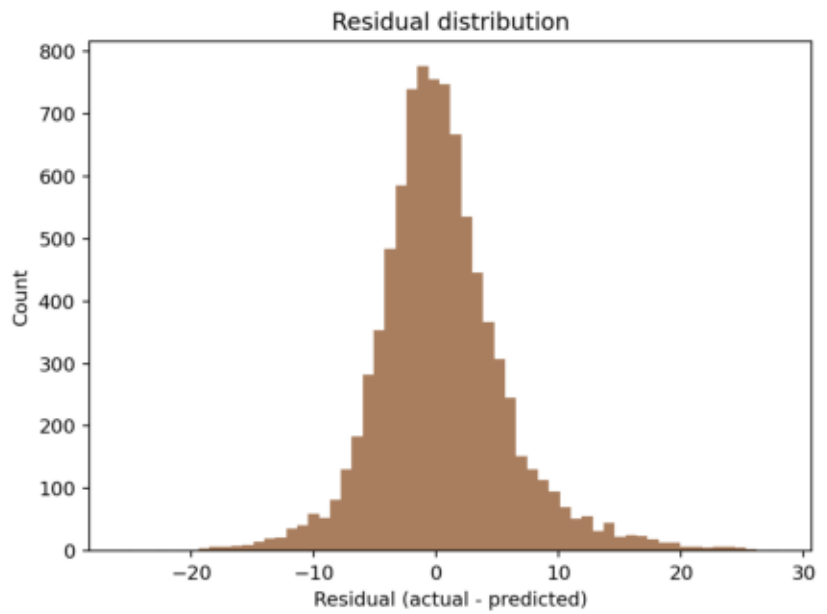


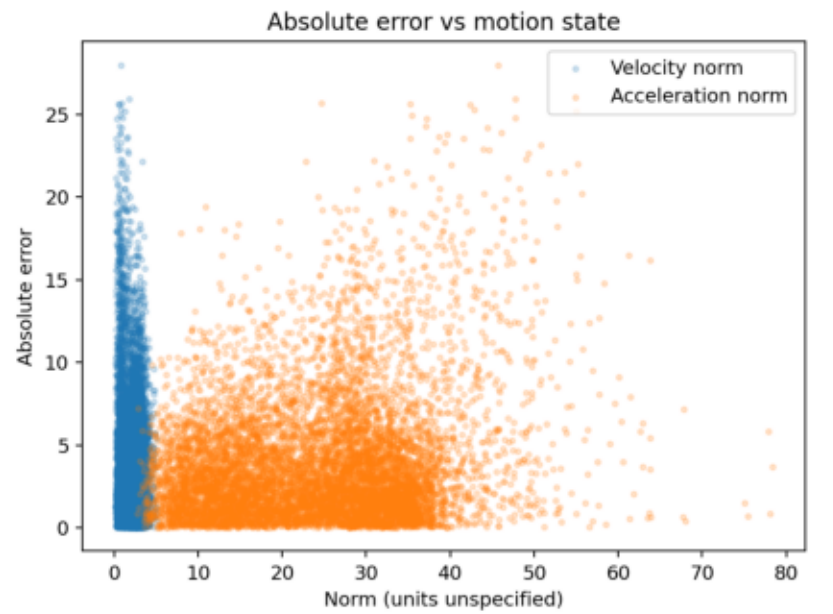
Blocked test residual diagnostics

Residual plots characterize the final Blocked Ridge fit on held-out rows. They are diagnostic descriptions, not evidence of episode-level transfer.



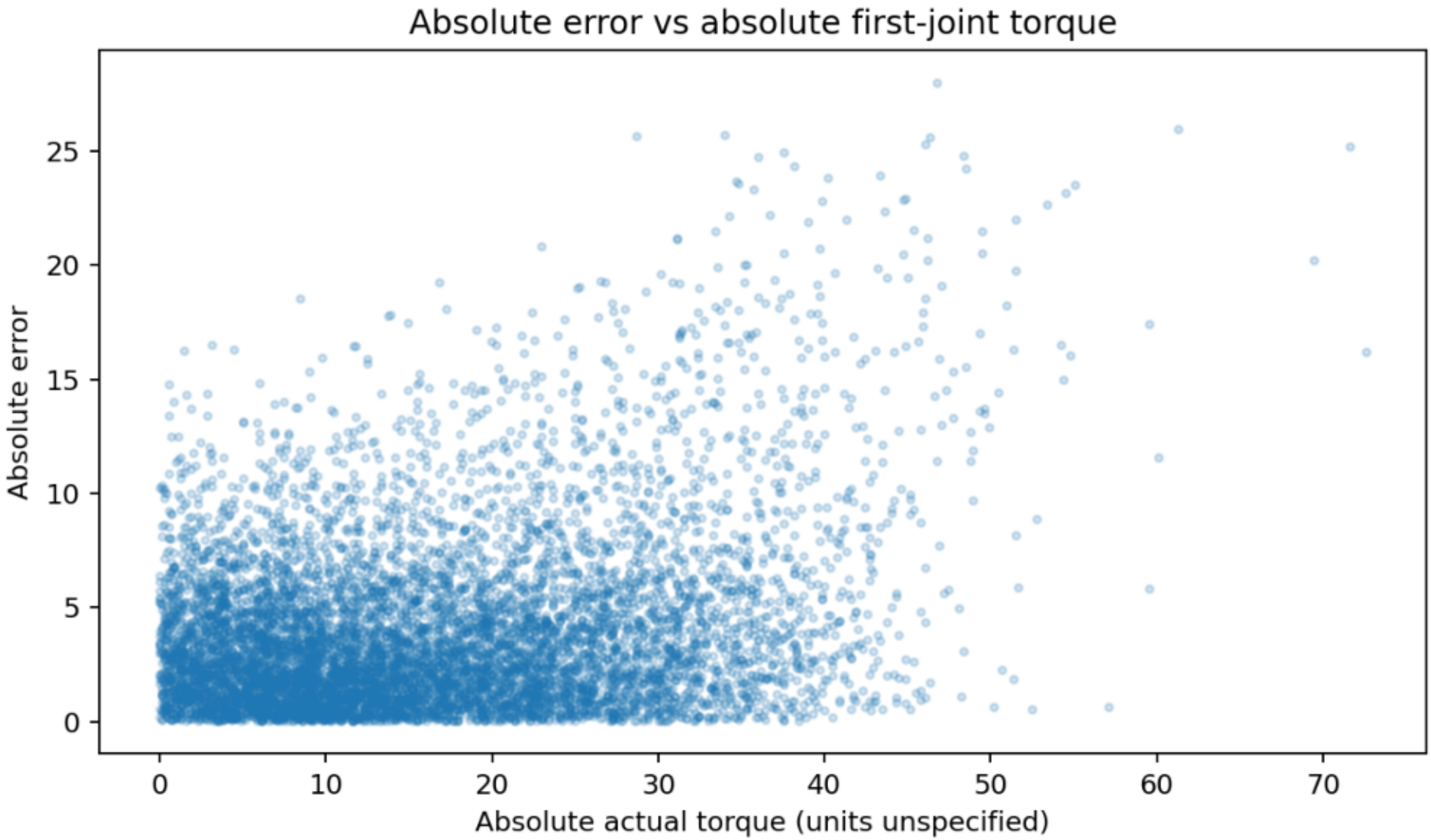






Absolute error versus absolute torque

The spread in absolute error increases in parts of the observed torque range. This plot does not normalize error by torque, so it cannot support claims about relative error.



Discussion, limitations, and conclusion

Conclusion: H1 is supported on the tested blocked evaluation. Ridge does not meaningfully improve on OLS under the tested models, alpha grid, dataset, and protocols. H3 is descriptive: high acceleration and high absolute torque slices have higher errors, while velocity error is non-monotonic.

Limitations: source-row dependence is substantial; timestamps and reliable episode identifiers are unavailable; the local 25-row purge does not establish independence. Blocked regions show distribution shift, and random partitions may retain nearby observations. The original test matrix overlaps the training matrix and is not an independent benchmark. No claim is made about new robots, tasks, or episodes.

The target's physical units and the dataset's redistribution license remain unverified. The public GitHub repository is the verified download source; the upstream GPML page is the dataset description and attribution reference. Further use or redistribution should follow the dataset owner's terms once verified.

References: Kaixhin/SARCOS, https://github.com/Kaixhin/SARCOS ; GPML dataset page, https://gaussianprocess.org/gpml/data/ ; split manifest: data/splits/sarcos_split_manifest.json ; protocol: docs/experiment_protocol.md.